Information and consent form

Energy and Amenity Data Collection in Healthcare Facilities in Malawi

Place and time:

Contact person if you have any questions: **Gian Teufen**, Global Health Engineering (GHE)
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Responsible District Health Office: **Rumphi DHO** (*ajumbe@medcol.mw*)

We would like to ask you if you are willing to participate in our research project. Your participation is voluntary. Please read the text below carefully and ask the conducting person about anything you do not understand or would like to know.

What is investigated and how?

This study addresses energy systems and the medical device and equipment inventory in healthcare facilities in Malawi. Data collection involves visits to approximately forty facilities throughout the country. The goal of this study is to create a public and freely accessible dataset of detailed energy system status and the medical device inventory functionality. This information will help us understand what medical devices are functional, how much energy the facility needs, what organization is responsible for the energy systems, and what works well and what doesn't.

Who can participate?

People at least 18 years old, who know about the energy system at their healthcare facility and have ideally worked there for more than two years.

What am I supposed to do as a participant?

You will be guided through a questionnaire about the facility. Then, we will ask you to show us the energy systems and some medical equipment. The investigators may also take photographs of energy systems or medical devices on the premises to document their condition.

What are my rights during participation?

Your participation in this study is voluntary. You can withdraw your participation at any time without giving reasons and without any disadvantage.

What risks and benefits can I expect?

The interview might take some of your time. However, your help will allow the government, aid groups, and planners to make better decisions and improve energy and resources for healthcare facilities in the future.

Will I be compensated for my participation?

No, we won't provide financial compensation.

What data is collected from me and how is it used?

The following data is collected from you:

- No personal data is collected from anyone.
- Data on the facility like amenities, devices, equipment, infrastructure, energy system and maintenance is collected and photos thereof will be taken.

Purpose of Data:

The data will be used for accurate modeling of energy needs of healthcare facilities, to understand the operations and maintenance of energy systems and in the future for scaling standalone solar microgrids for rapid facility electrification. Further, we'll use data on facility amenities and equipment for an estimation of the service readiness indicators and index as part of the SARA evaluation tool by WHO. This will provide an up-to-date nationwide data set on the status of small healthcare facilities.

Handling of Identifying Data:

We will not collect any personal information that can identify you, so you will remain completely anonymous. This paper consent form with your name on it will be stored in a locked cabinet at ETH Zurich for five years, accessible only by Prof. Dr. Tilley.

Publication

Our findings will be shared in research papers and easier-to-read formats like blogs and reports. No one will be able to identify you from anything we publish. The anonymous dataset will be published openly for further use as an R package and a public GitHub repository for analysis code allowing other researchers to build upon our findings.

The collected data of your facility and the full national analysis will be shared with the facility administrator and the corresponding District Health Office (DHO) once digitized.

Who is funding the study?

The study is funded by the Global Health Engineering (GHE) group at ETH Zurich, Switzerland

Who reviewed the study?

This study was reviewed and approved by the Mzimba District Health Office and the ETH Zurich Ethics Committee under application number 25 ETHICS-206.

Complaints office

The secretariat of the ETH Zurich Ethics Committee is available to help you with complaints in connection with your participation in the study. Contact: ethics@sl.ethz.ch or +41 44 632 85 72.

Consent Form

I, the participant, confirm with my signature that:

- I have read and understood the study information.
- I had enough time to decide about my participation and am taking part in the study voluntarily.
- I fulfil the stated conditions for participation.
- I consent that the data described above may be collected from me and used as described.
- I know that the data collected from me may be published in anonymized form.
- I know that I can cancel my participation any time.

I would like to be informed about the results of this study

☐ Yes. Name, phone number or e-mail: ……

☐ No.

Name of participant

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Place, date

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Signature participant

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Place, date

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Signature conducting person

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